PSP Active Controls



Title: *Regenerative Cooling System with Calorimeter Experiment for Thermal Model Verification*

Presentors: *Pavit Hooda & Grant Williams*

A team dedicated to vertical take-off and vertical landing technology.

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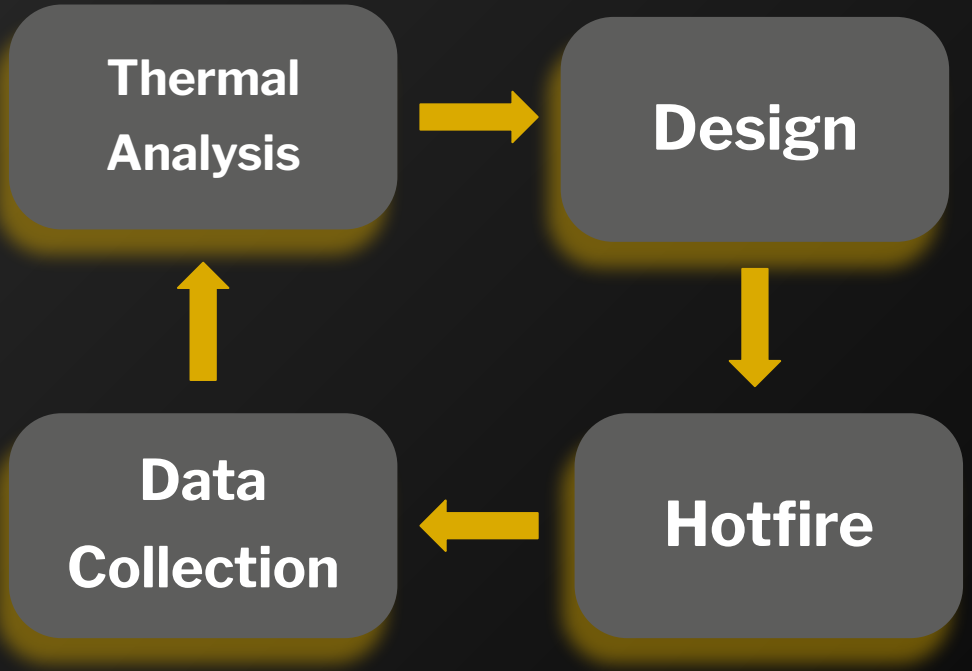


Proposition

To experimentally validate the thermal model of regenerative cooling channels with the second ever hotfire of an additively manufactured AL6061 test chamber article.



Method

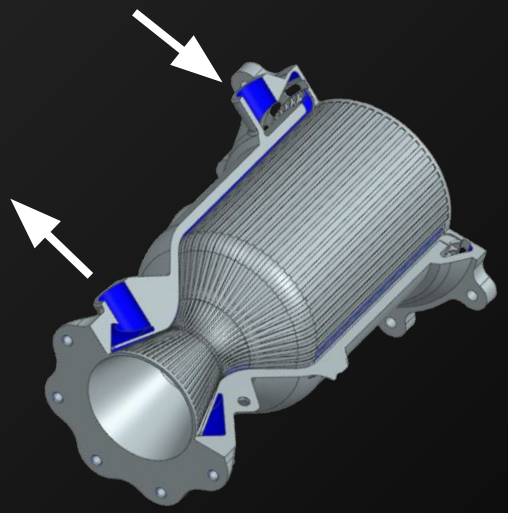


Design

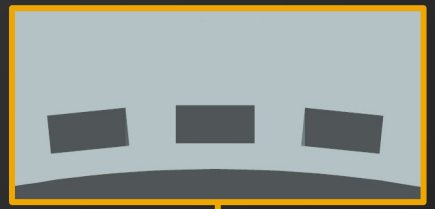
62 cooling channels

Water flow rate of 4 lb/s - 7 lb/s

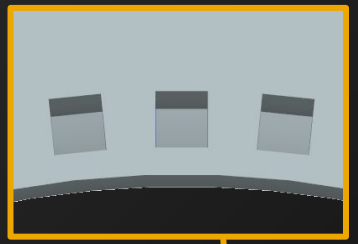
400 °F designed wall temperature



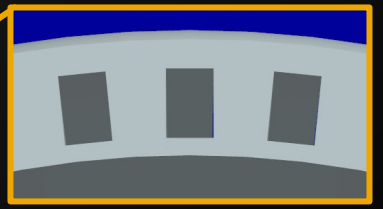
Chamber
0.059 x 0.122“



Throat
0.039 x 0.039“



Exit
0.091 x 0.059“

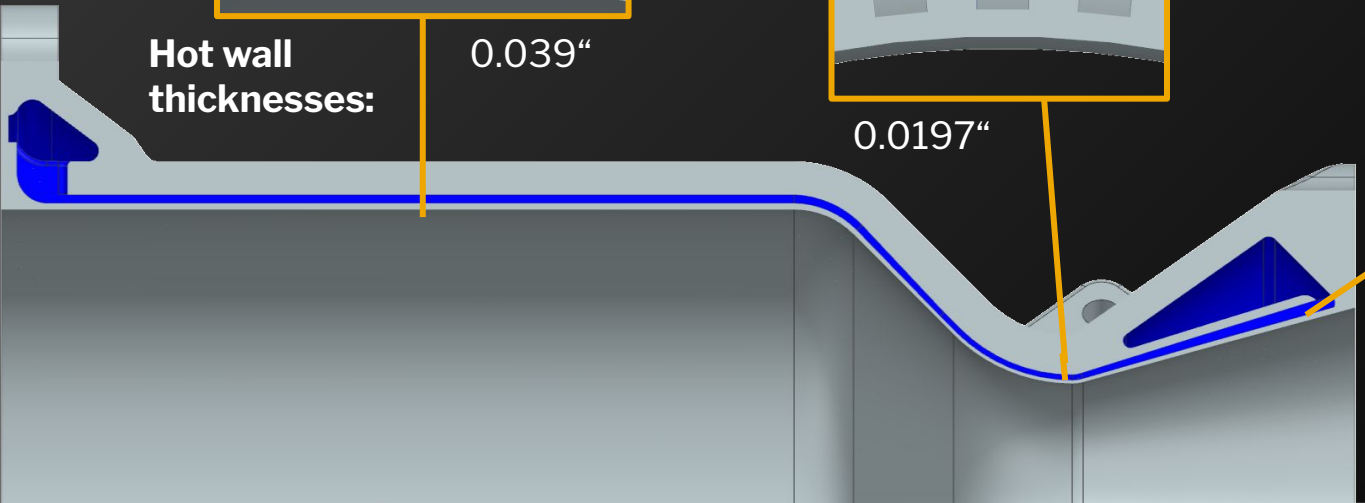


Hot wall thicknesses:

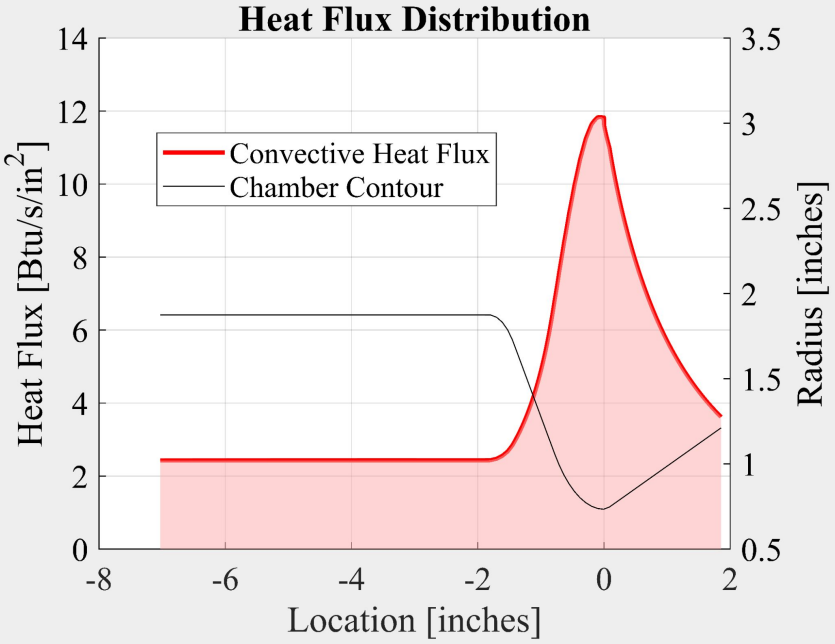
0.039“

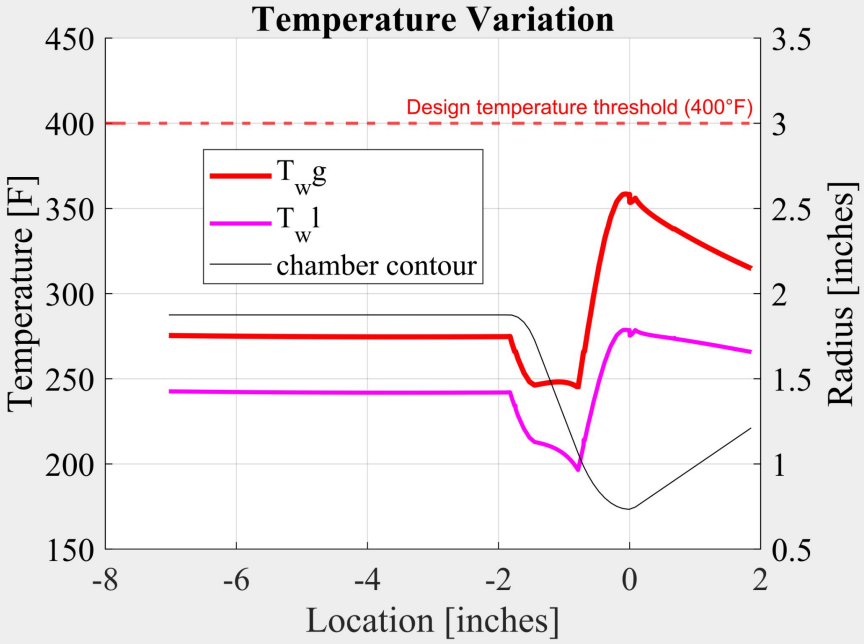
0.0197“

0.039“



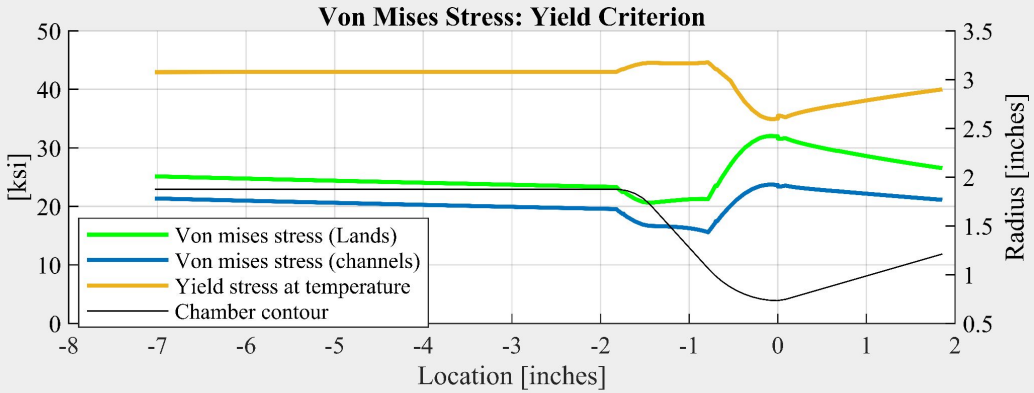
Analytical Thermal Analysis

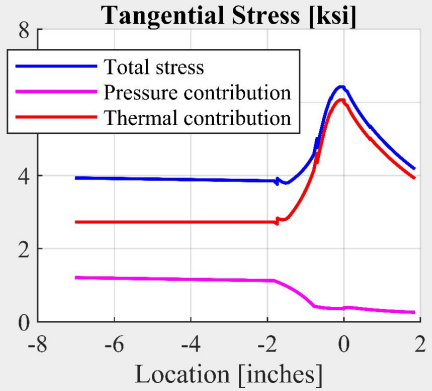


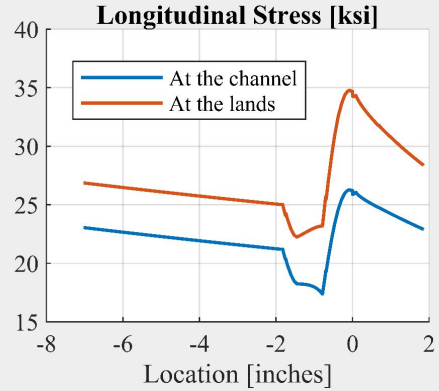


Bulk Specific Heat Flux: 1.680×10^6 BTU/lbs

Analytical Stress Analysis

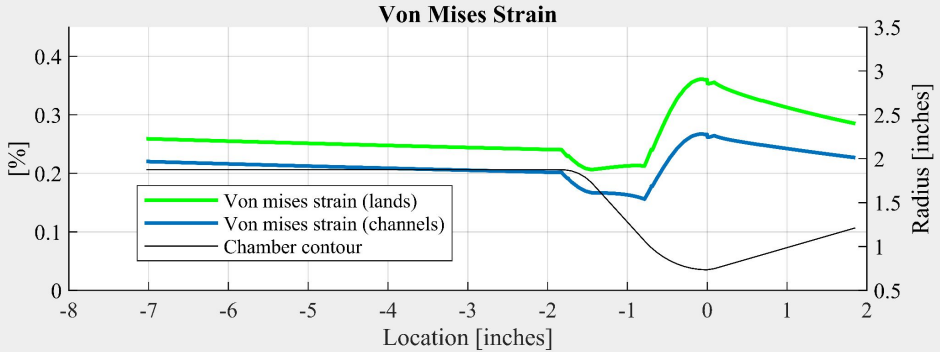


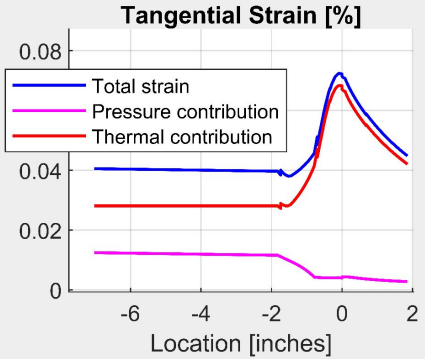


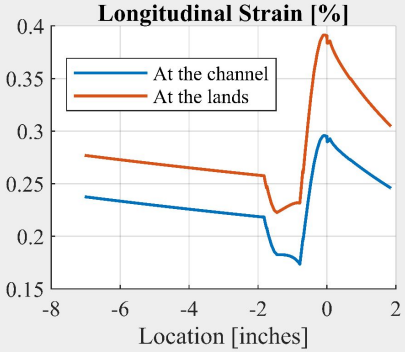


- FOS of **1.09** to yield via Von Mises failure criterion
- Uses temperature-varying properties for calculations
- Load sources: thermal expansion, engine pressure, coolant pressure

Analytical Strain Analysis







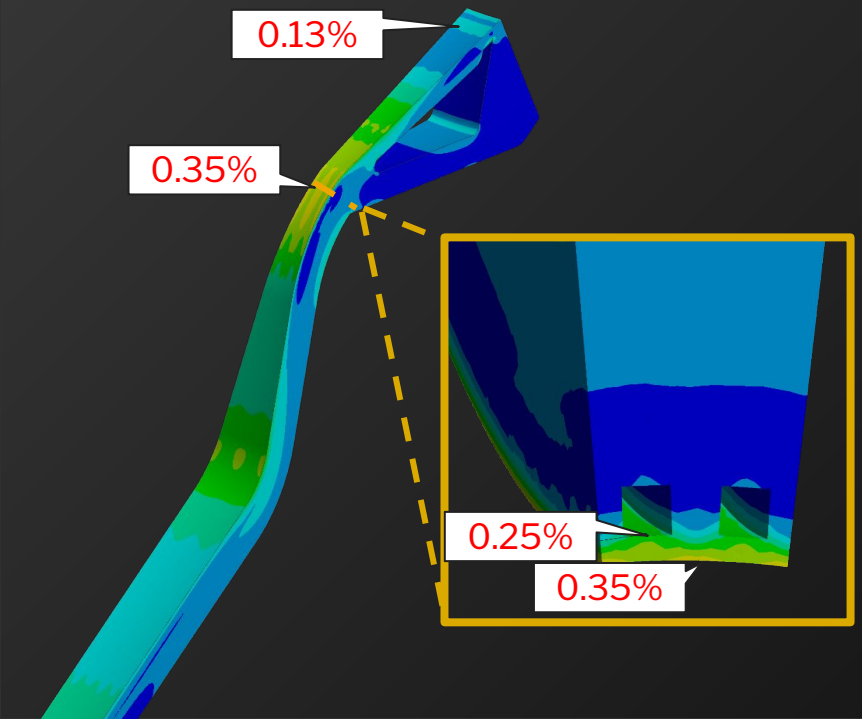
Strain	Channel	Lands
Equivalent	.26%	.36%
Total effective	.44%	

$$\epsilon_t = \frac{1}{E}\frac{(p_l - p_g)}{2}\left(\frac{w}{t_w}\right)^2 + \alpha\Delta T_1 \qquad \epsilon_l = \alpha(\Delta T_1 + \Delta T_2)$$

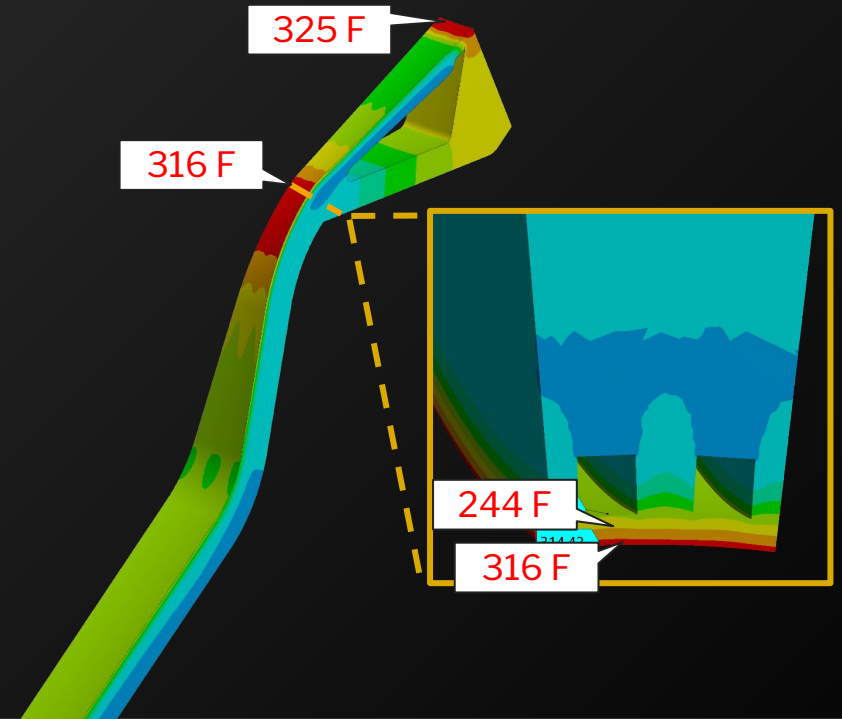
ϵ_t	Tangential Strain	w	Channel Width	t_w	Wall Thickness
ϵ_l	Longitudinal Strain	ΔT_1	Hot wall temp difference	E	Young's Modulus
p_l	Coolant Pressure	p_g	Chamber Pressure		
ΔT_2	Hot wall and bulk coolant temp difference	α	CTE		

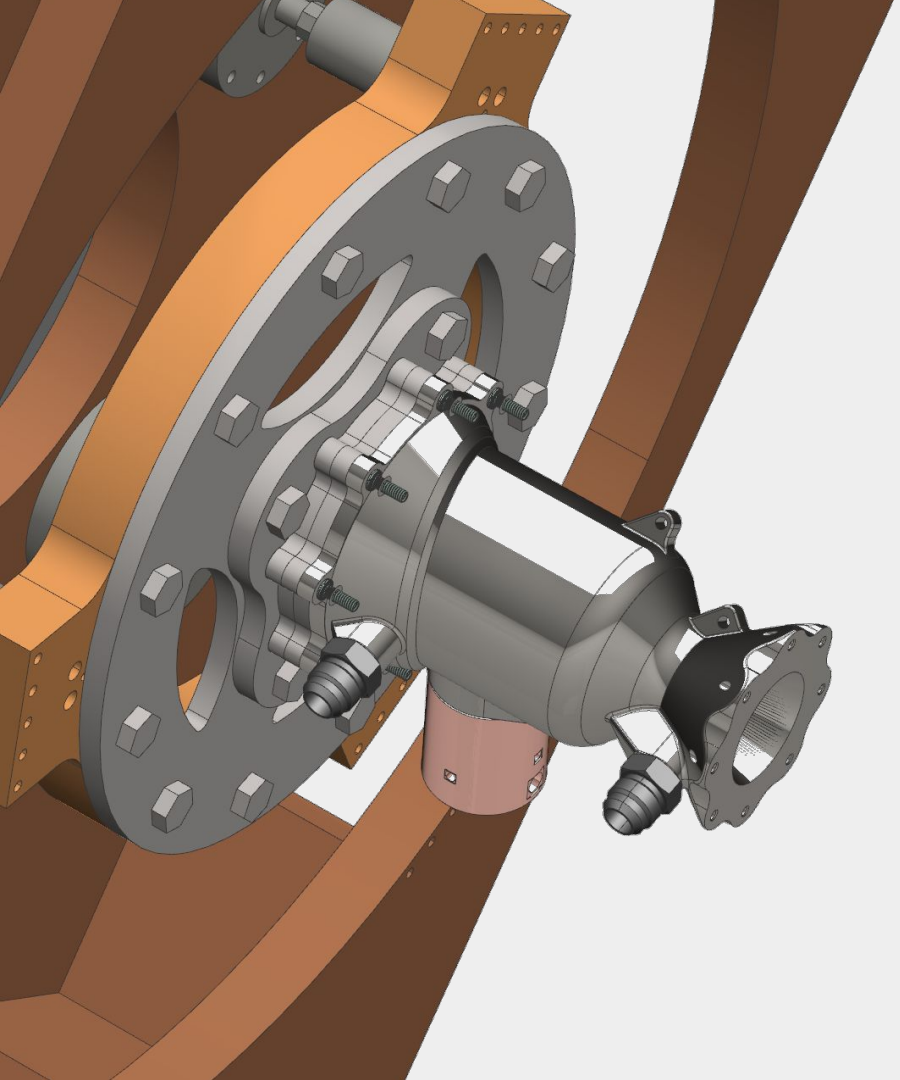
FEA Results

Equivalent Elastic Strain [%]



Temperature [°F]



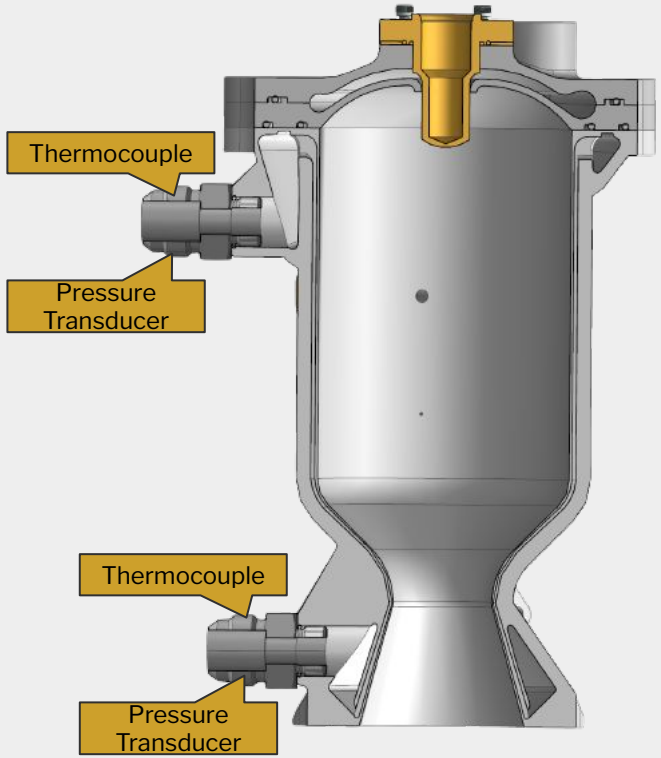


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First hotfire February, 2024!

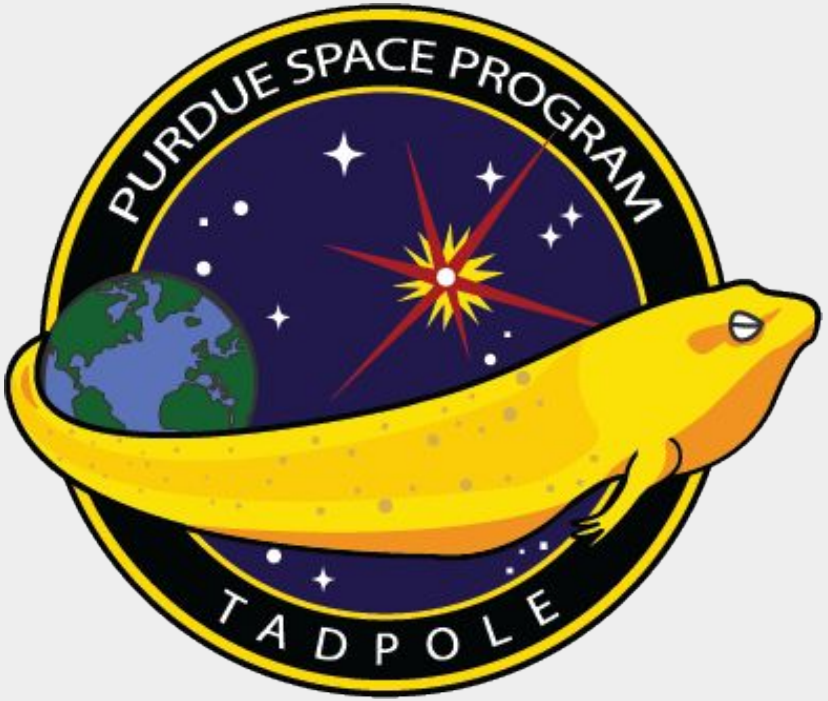
Pre-Conclusion

Test #	Description	Objectives
1-2	Ignition Validation	Verify ignition timings, injector stability, measure full startup and shutdown transients
3	Full Throttle	Verify full power level performance and TCA structural integrity, heat flux, injector stability
4	Minimum Throttle	Verify lowest throttle level performance, injector stability at low dP/Pc, low throttle level heat flux
5	Gimbal Hotfire	CPLC milestone 1: gimbal 7 degrees in a circle
6	Throttle Hotfire	CPLC milestone 2: throttle smoothly down to 40%
7-11	Gimbal Refinement	Trim and verify gimbal parameters
12-16	Throttle Refinement	Trim and verify throttle parameters
17-20	Regen Testing	Repurpose water jacket for flight-like regen



Thank You

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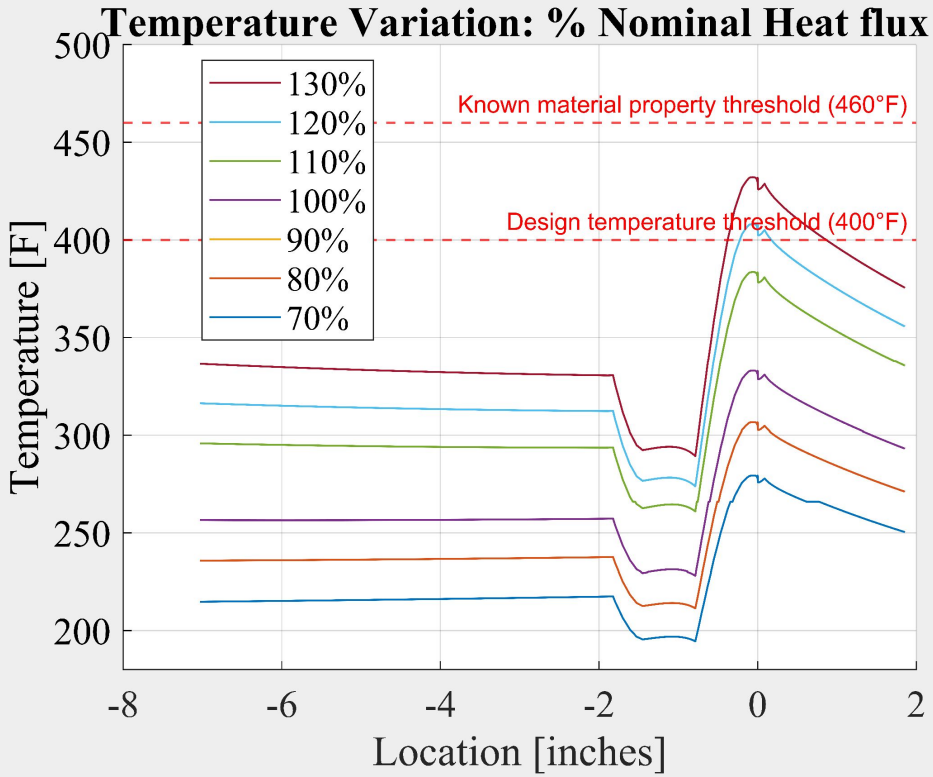
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Sources Cited

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2. Huzel, D. K., & Huang, D. H. (1992). *Modern Engineering for design of liquid-propellant rocket engines*. American Inst. of Aeronautics and Astronautics.
3. Bartz, D. R. (1957). A Simple Equation for Rapid Estimation of Rocket Nozzle Convective Heat Transfer Coefficients. *Jet Propulsion*, *27*, 49–51.
4. (1972). (tech.). *Liquid Rocket Engine Fluid-Cooled Combustion Chambers* (Vol. SP-8087, Ser. NASA Space Vehicle Design Criteria (Chemical Propulsion)).
5. Kuyper, D. J., & Burge, H. L. (1967). Simplified Thermal Fatigue Analysis for Liquid Rocket Combustion Chambers. *J. Spacecraft Rockets, 4*, 126–128.

Temperature Variation



Instrumentation

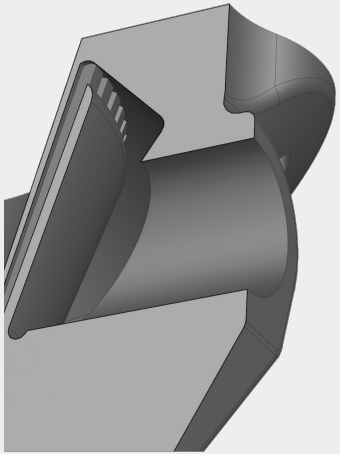
- Plan to use existing sensors from Zucrow
  - Will purchase additional sensors pending Zucrow availability

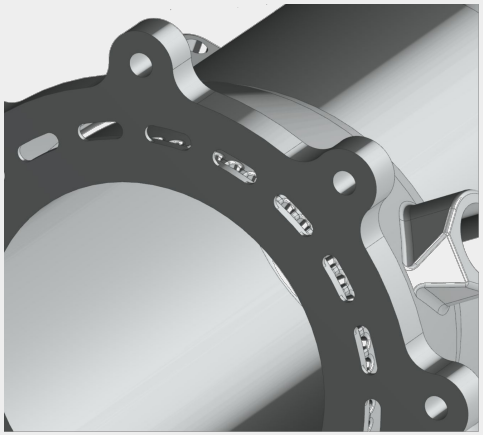
Sensor	Family
PT	UNIK 5000
TC	Omega K Type

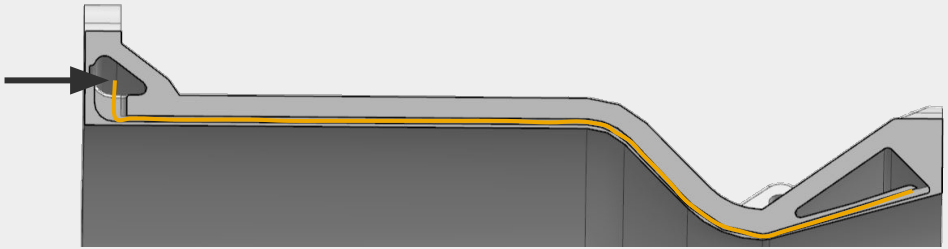


Depowdering

- Initial depowdering done by Elementum
- Channel access holes at injector seal for further cleaning with nichrome wire in case of blockage / after post machining
- Will use a CAT scan or X-RAY to ensure no channels are blocked



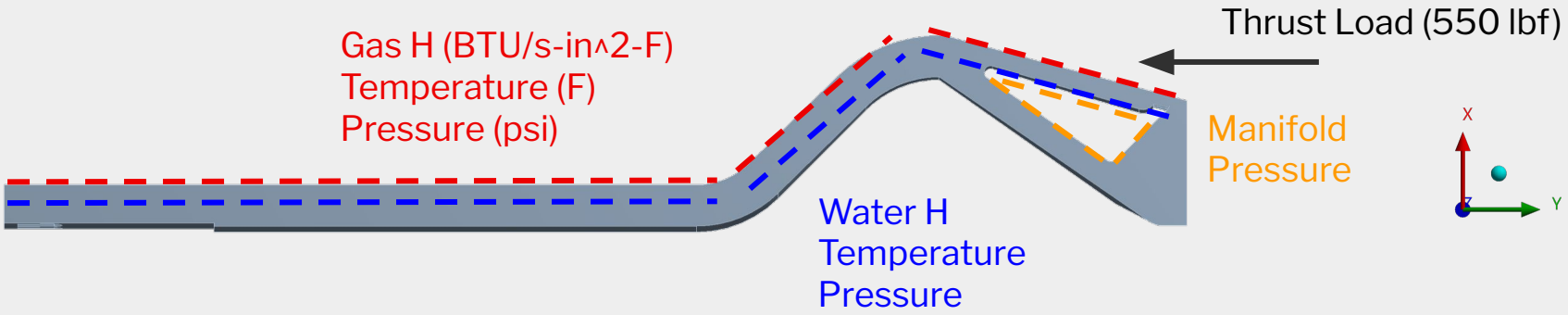




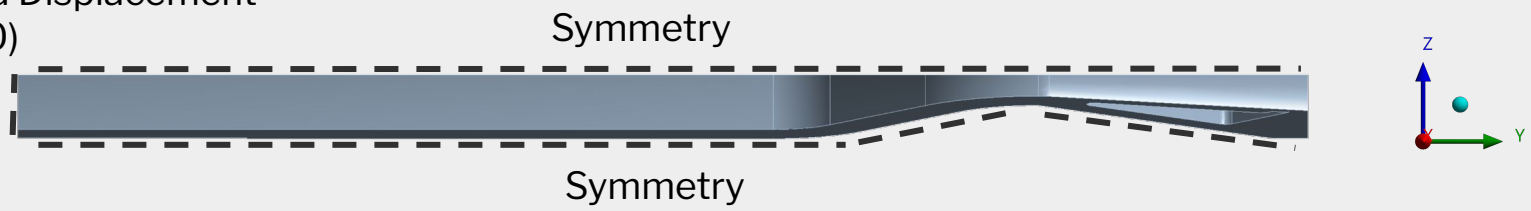
Thermo-Structural FEA Setup

3D FEA of 2 channels

Spatially varying boundary conditions



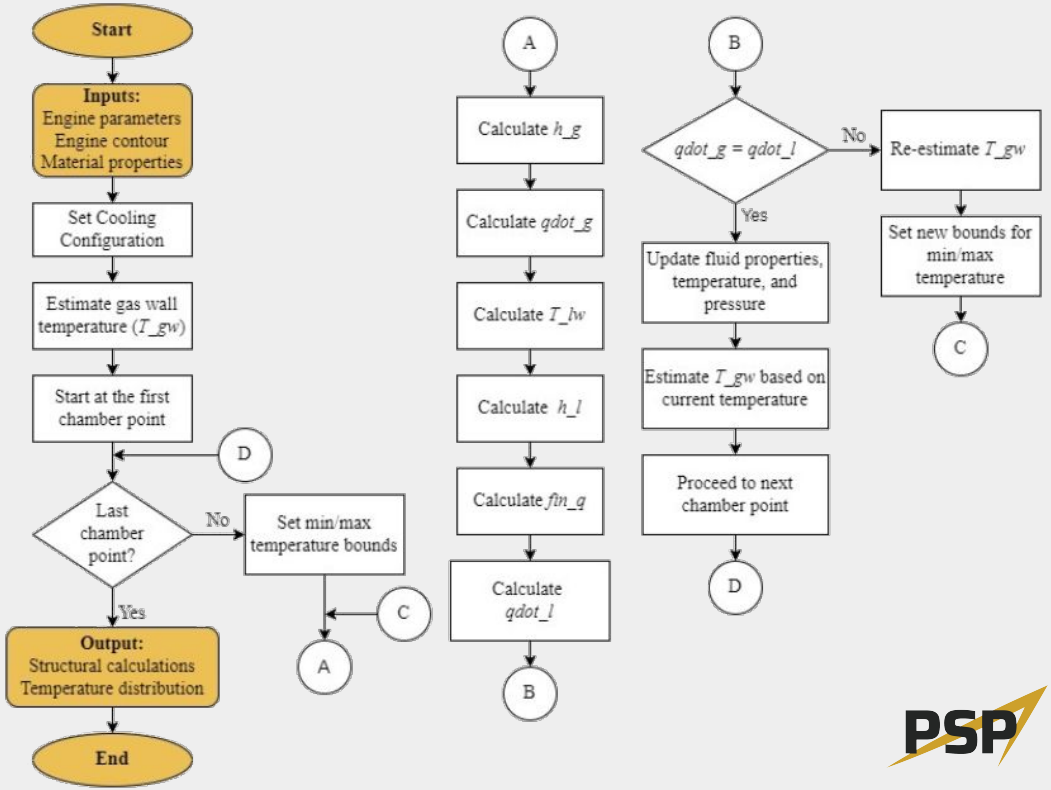
Fixed Displacement
(y = 0)



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Heat Transfer Model

- Assumptions: steady state, no backside heat transfer, equally distributed temp in channels, adiabatic fin tip, no radiation
- NASA CEA for combustion properties (Frozen)
- Coolprop for coolant properties
- Changing material properties with temperature
- Bartz and Nusselt correlations for heat transfer



Thermal-Structural Analysis Outline

- Low cycle fatigue analysis to determine chamber life using worst-case results

